

18.06 Problem Set 5

Due Friday, October 14 at 11am.

Problem 1 (5+5+5)

Any linearly independent vectors form a basis, but some basis choices can have distorting effects on data. In particular, *nearly* parallel vectors can have unpleasant results as a basis.

Consider the vector space $\mathbb{R}^2$ and the *nearly parallel* basis vectors $\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\vec{a}_2 = \begin{pmatrix} 1 \\ 10^{-6} \end{pmatrix}$.

(a) Write the vectors $\vec{b} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \vec{a}_1 x_1 + \vec{a}_2 x_2$ and $\vec{c} = \begin{pmatrix} 1 \\ 0.1 \end{pmatrix} = \vec{a}_1 y_1 + \vec{a}_2 y_2$ in the basis $\vec{a}_1, \vec{a}_2$: find the coefficients $\vec{x}$ and $\vec{y}$.

(b) Compare $\|\vec{c} - \vec{b}\|$ to $\|\vec{y} - \vec{x}\|$: what is the ratio $\|\vec{y} - \vec{x}\|/\|\vec{c} - \vec{b}\|$ (to 2 significant digits)?

(c) Suppose that you instead have an *orthonormal* basis given by the columns of Q (an $m \times n$ matrix ... not just $\mathbb{R}^2$ in this part). For *any* vectors $\vec{b} = Q\vec{x}$ and $\vec{c} = Q\vec{y}$ in this basis, show that $\|\vec{y} - \vec{x}\| = \|\vec{c} - \vec{b}\|$: orthonormal bases "preserve distances".

Problem 2 (5+5+5)

Suppose we have four unknowns $\vec{x} = [x_1, x_2, x_3, x_4] \in \mathbb{R}^4$, and we want to solve the four *nonlinear* equations:

$$\underbrace{\begin{pmatrix} -5 & -1 & -2 \\ -1 & -2 & 4 \\ -2 & 4 & -8 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_4 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix},$$
$$x_1^2 + x_2^2 + x_3^2 = 1.$$

(a) Rewrite this equation in the form $\vec{f}(\vec{x}) = \vec{0}$, and compute both $\vec{f}(\vec{x})$ and its Jacobian matrix $J(\vec{x})$.

(b) Fill in the Julia functions below to compute f and J , and check that $\vec{f}(\vec{x} + \vec{\delta}) - \vec{f}(\vec{x}) \approx J(\vec{x})\vec{\delta}$ for a $x = [1, 2, 3, 4]$ and a "random" small vector $\delta = [-2, 3, 4, 1] \times 10^{-8}$, using the sample code below.

(c) Starting with $x = [1, 2, 3, 4]$, run a few steps of Newton's method (either copy-and-paste a few steps `@show x = x - J(x) \ f(x)` by hand, or write a loop), and verify that it converges quickly. Check that after a few steps you have a very accurate solution: make sure that `A*x[1:3]` is nearly `x[4]*x[1:3]` and that `x[1:3]'*x[1:3]` is nearly `1`. How many Newton steps did it take for `x[4]` to have 4 correct decimal digits?

```
In [ ]: # part (b)

using LinearAlgebra

A = [ -5  -1  -2
      -1  -2   4
      -2   4  -8]

function f(x)
    return ??? # return a 4-component vector
end

function J(x)
    return ??? # return a 4x4 matrix of the Jacobian
end
```

```
In [ ]: # part (b) continued: check

x = [1,2,3,4]
δ = [-2,3,4,1] * 1e-8 # a "random" small vector

# these should be very close in value:
@show f(x+δ) - f(x)
@show J(x)*δ
```

```
In [ ]: # part (c)

x = [1,2,3,4]

# a single Newton step:
@show x = x - J(x) \ f(x)
```

```
In [ ]: # part (c): checks

@show f(x)

@show A*x[1:3]
@show x[4]*x[1:3]
@show x[1:3]'*x[1:3] # same as x[1]^2 + x[2]^2 + x[3]^2

# Was x a correct solution?
```

Problem 3 (5+5+10)

In class, we showed how to differentiate some matrix-valued functions:

$d(A^2) = AdA + dAA$, which is a linear operator acting on dA , and also

$d(A^{-1}) = -A^{-1}dAA^{-1}$.

(a) Write $d(A^4)$ as a linear operator acting on dA .

(b) We can compute $f(A) = x^T A^{-1}y$ (a scalar-valued function of a square invertible matrix A) by solving only *one* linear system $z = A^{-1}y$ (i.e. solving $Az = y$) and then taking a dot product. Show that df (also a scalar!), for any infinitesimal change dA , can be computed by solving only *one* additional linear system _____ = _____ (in addition to solving $Az = y$), and then taking some matrix/vector products. (This is called an "adjoint solve" by engineers or "backpropagation" by computer scientists.)

(c) Let $A = \begin{pmatrix} a_1 & a_3 \\ a_2 & a_4 \end{pmatrix}$ be a 2×2 matrix. Let $\text{vec}(A) = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \vec{a}$ be the vector

formed by "stacking" the columns of A (this is computed by `vec(A)` in Julia).

Equivalently, $\vec{a}$ are the coefficients you get from expanding A in the basis

_____. Let $dA = \begin{pmatrix} da_1 & da_3 \\ da_2 & da_4 \end{pmatrix}$. Show that

$\text{vec}(AdA + dAA) = (\text{some matrix})\text{vec}(dA)$: find the 4×4 "some matrix" (which expresses the linear operator $AdA + dAA$ in this "vectorized" basis). Check your answer in Julia for an arbitrary choice of `A` and `dA`:

```
In [ ]: # check your answer for part (c):

A = [1 2
      3 4]
a = vec(A)
dA = [7 9
       6 -3]

some_matrix = ????? # your answer from (c)

vec(A*dA + dA*A) ≈ some_matrix * vec(dA) # should return "true"
```

Problem 4 (5+5+5+5)

(From Strang, section 4.4.)

- (a) If A has three orthogonal columns of lengths 1,2,3, what is $A^T A$?
- (b) Give an example of a matrix Q with orthonormal columns but $QQ^T \neq I$.
- (c) Give an example of two orthogonal vectors that are *not* linearly independent.

- (d) If we have a basis $a_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, a_2, a_3 of orthogonal vectors in $\mathbb{R}^3$ that are *not* normalized to length 1, and

$$\underbrace{\begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}}_b = Ax = a_1 x_1 + a_2 x_2 + a_3 x_3,$$

then $x_1 =$ _____.